

JOINT ONE-SIDED TESTS OF LINEAR REGRESSION COEFFICIENTS*

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The main example of the class of problems considered below is that of testing whether a subset of regression coefficients are jointly zero assuming knowledge of the coefficients' signs. If this knowledge is ignored, the likelihood ratio, Wald, and Lagrange multiplier tests are each equivalent to the F -test. We propose a new test which can be applied as a one-sided t -test and which is UMPI in a subspace of the parameter space. Empirical power comparisons with the power envelope, the F -test, and the exact one-sided likelihood ratio test show that the new test can have exceptionally good power over a wide range of the parameter space.

1. Introduction

In recent years there has been a heavy emphasis on the use of three classical test procedures in econometrics. These are the likelihood ratio, Wald, and Lagrange multiplier test procedures. They are generally applied as two-sided tests in order to take advantage of the convenient fact that, subject to regularity conditions, the test statistic has an asymptotic chi-squared distribution under the null hypothesis. As King (1986) points out, there is a fundamental reason why such test procedures are not totally suited to econometric applications. It is that most econometric testing problems are potentially one-sided because economic theory is usually rather good at providing information about the signs of parameters. This is particularly true of linear regression coefficients. As we shall see below,¹ ignoring information about signs of parameters can lead to a considerable loss of power in small samples.

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¹Also see King and Evans (1984).

This paper considers the familiar problem of testing a set of joint restrictions on the coefficients of the standard linear regression model. The restrictions are assumed to be at the boundary of a set of joint inequality restrictions that economic theory imposes upon the coefficients. For example, we may wish to test the hypothesis that two regression coefficients are jointly zero given that economic considerations imply that they cannot be negative.

The standard textbook solution to this problem is to ignore the inequality restrictions under the alternative hypothesis and apply the usual F -test. As is well-known, this test is equivalent to the likelihood ratio, Wald, and Lagrange multiplier tests for the unconstrained problem. However, because it ignores the inequality restrictions under the alternative hypothesis, it is clearly not the most sensible test to apply to this problem.

Recently, Gouriéroux, Holly and Monfort (1982) discussed one-sided versions of the likelihood ratio, Wald, and Lagrange multiplier tests for this problem. They call their one-sided version of the latter test the Kuhn–Tucker test. They showed that the asymptotic distribution of each of the test statistics under the null hypothesis is a weighted sum of chi-squared distributions and the degenerate distribution at zero. Because of this, when there is one restriction their tests are not equivalent to the familiar one-sided t -test at significance levels of higher than² 0.5. This is of interest because the one-sided t -test is known to be Uniformly Most Powerful Invariant (UMPI). The exact distribution of the likelihood ratio test in the case of two restrictions is derived by Hillier (1986) who uses the technique of conditioning on a sufficient statistic. Mention should also be made of Yancey, Judge and Bock (1981) and Yancey, Bohrer and Judge (1982) who study a related testing problem but in the restricted context of the orthonormal linear regression model.

Obviously for any testing problem we would like to use a uniformly most powerful test. The power function of such a test is equal to the envelope of maximum obtainable power. As is generally the case, for our problem such a test does not exist. The next best option is to use a test whose power is very close to the power envelope over that part of the parameter space that is of interest. It would be an added bonus if the test were easy to apply.

In this paper we present a test which conforms closely to this second best option for a wide class of testing situations likely to occur in practice. The method of test construction found by King (1983a, 1983b, 1984, 1985) to be highly successful for problems involving the covariance matrix of regression disturbances is used here to derive a test which is UMPI in the neighbourhood of a subspace of the parameter space. The resultant test can be applied as a one-sided t -test of a regression coefficient. It can also be used to obtain the

²Such a distinction may not seem important, but occasionally in pre-testing situations, it may be desirable to apply the test with a high significance level. See, for example, the discussion in Fomby and Guilkey (1978) and King and Giles (1984).

maximum invariant power envelope for any alternative hypothesis model. Section 3 reports the results of an empirical power comparison of the power envelope with the small sample power of the new test, the F -test and, in the case of two restrictions, the likelihood ratio test. Some concluding remarks are made in the final section.

2. Theory

Consider the standard linear regression model

$$y = X\beta + u, \quad (2.1)$$

where y is $n \times 1$, X is an $n \times k$ non-stochastic matrix of rank $k < n$, β is a $k \times 1$ vector of unknown parameters, and u is an $n \times 1$ disturbance vector. Further, it is assumed that

$$u \sim N(0, \sigma^2 I_n), \quad (2.2)$$

where σ^2 is unknown.

We are interested in testing the null hypothesis,

$$H_0: R\beta = r,$$

against the alternative,³

$$H_a: R\beta \geq r \quad (\text{excluding } R\beta = r), \quad (2.3)$$

where R is a known $j \times k$ matrix of rank $j < k$, and r is a known $j \times 1$ vector. Without loss of generality we will assume that

$$R = [0: I_j] \quad \text{and} \quad r = 0,$$

because if this is not the case, (2.1) and (2.3) can be transformed to

$$y^* = X^*\beta^* + u \quad \text{and} \quad [0: I_j]\beta^* \geq 0,$$

respectively, where

$$y^* = y - X^*[0': r']',$$

$$X^* = XT^{-1},$$

$$\beta^* = T\beta - [0': r']',$$

³If a and b are vectors of the same dimension then $a \geq b$ denotes $a_i \geq b_i$ for every i .

and T is any $k \times k$ non-singular matrix such that

$$RT^{-1} = [0: I_j].$$

In other words, we wish to test the null hypothesis that the last j components of β are zero against the alternative that one or more are positive.

Partition (2.1) accordingly, that is with $l+j=k$ write

$$y = X_1\beta_1 + X_2\beta_2 + u, \quad (2.4)$$

where $X = [X_1: X_2]$ and $\beta' = (\beta_1': \beta_2')$ are such that X_1 is $n \times l$, X_2 is $n \times j$, β_1 is $l \times 1$ and β_2 is $j \times 1$. Our problem is now one of testing

$$H_0: \beta_2 = 0,$$

against

$$H_a: \beta_2 \geq 0 \quad (\text{excluding } \beta_2 = 0).$$

First, consider the simpler problem of testing $H_0: \alpha = 0$ against $H_a: \alpha > 0$ with respect to (2.4), where $\beta_2 = \alpha\beta_2^*$, α is an unknown scalar, and β_2^* is a known and fixed $j \times 1$ vector. Because $z = X_2\beta_2^*$ is non-stochastic, it follows that the standard one-sided t -test on α in

$$y = X_1\beta + \alpha z + v \quad (2.5)$$

is UMPI against H_a , where invariance is with respect to transformations of the form

$$\bar{y} = \gamma_0 y + X_1\gamma,$$

in which γ_0 is a positive scalar and γ is an $l \times 1$ vector. For the wider problem of testing H_0 against H_a , this test is UMPI in the neighbourhood of

$$\beta_2 = \alpha\beta_2^*, \quad \alpha > 0. \quad (2.6)$$

If a test that is UMPI against $\beta_2 \geq 0$ does exist, it would be equivalent to this test. However, because this test is clearly not independent of the choice of β_2^* , no UMPI test exists for the wider problem.

Using the formula for partitioned inverses, the ordinary least squares estimator for α in (2.5) can be shown to be

$$\hat{\alpha} = (z' \bar{P}_1 z)^{-1} z' \bar{P}_1 y,$$

where

$$\bar{P}_1 = I_n - X_1(X_1'X_1)^{-1}X_1'.$$

Under (2.1) and (2.2), and noting that $\bar{P}_1X = [0 : \bar{P}_1X_2]$,

$$(z'\bar{P}_1z)^{1/2}\hat{\alpha}/\sigma \sim N(\zeta, 1),$$

where

$$\zeta = (z'\bar{P}_1z)^{-1/2}z'\bar{P}_1X_2\beta_2/\sigma.$$

Let W denote the $n \times (l+1)$ matrix

$$W = [X_1 : z],$$

let $\hat{v}$ denote the ordinary least squares residuals from (2.5), and let $s^2 = \hat{v}'\hat{v}/(n-l-1)$ be the usual unbiased estimator of σ^2 . Note that under (2.1) and (2.2),

$$\hat{v}'\hat{v}/\sigma^2 = (n-l-1)s^2/\sigma^2 \sim \chi'^2(n-l-1; \lambda),$$

where $\chi'^2(n-l-1; \lambda)$ is the non-central chi-squared distribution with $n-l-1$ degrees of freedom and non-centrality parameter

$$\lambda = \beta_2'X_2'\bar{P}_wX_2\beta_2/\sigma^2,$$

in which

$$\bar{P}_w = I_n - W(W'W)^{-1}W'.$$

The t -statistic for a test of $\alpha = 0$ in (2.5) is

$$\begin{aligned} t &= \hat{\alpha} / \left\{ s(z'\bar{P}_1z)^{-1/2} \right\} \\ &= \left\{ (z'\bar{P}_1z)^{1/2}\hat{\alpha}/\sigma \right\} / \left[\hat{v}'\hat{v} / \{ \sigma^2(n-l-1) \} \right]^{1/2}. \end{aligned}$$

As both $\hat{\alpha}$ and $\hat{v}$ are independent by construction, it follows that under (2.1) and (2.2)

$$t \sim t''(n-l-1; \zeta, \lambda),$$

where $t''(n-l-1; \zeta, \lambda)$ is the doubly non-central t -distribution with $n-l-1$ degrees of freedom and numerator and denominator non-centrality

parameters ζ and λ , respectively. [See for example Krishnan (1968).] Under H_0 , $\zeta = \lambda = 0$ and the test statistic has a central t -distribution with $n - l - 1$ degrees of freedom.

Before the test can be applied, a value for β_2^* must be decided upon. This must be done without reference to y otherwise the test may not have the correct size and will not be UMPI in the neighbourhood of (2.6). The test is then applied as a simple one-sided t -test of $\alpha = 0$ in (2.5) where z is constructed as a weighted sum (with the components of β_2^* as weights) of the X_2 regressors. A researcher most likely would wish to set β_2^* using his prior knowledge about the relative magnitudes of the β_2 coefficients. This has the advantage that if the researcher changes the scale of any of the X_2 regressors, β_2^* would change appropriately and the outcome of the test would be unaffected. In the absence of such prior knowledge, another obvious approach is to set $\beta_2^* = (1, 1, \dots, 1)'$ which is the choice of β_2^* value we will investigate in the remainder of this paper. Because this choice of β_2^* implies that the test reduces to a standard one-sided t -test on the coefficients of the sum of the X_2 regressors in (2.5), we will call this test the additive t -test.

A weakness of this choice of β_2^* is that it will result in a different test if the scales of some but not all of the X_2 regressors are changed. This is because $z = X_2 \beta_2^*$ is not invariant to such scale changes while $X_2 \beta_2$ is. In many applications it will be natural for all the X_2 regressors to have the same scale, so this will not be an issue. A number of examples of such applications may be found in the empirical power comparison reported in the following section. When it is an issue, one needs to ensure that the scales of the X_2 regressors are chosen sensibly before the test is applied.

It may be that in some cases the test is not particularly sensitive to the choice of β_2^* value. For any given testing problem, one can trace out the power envelope by varying β_2^* and calculating the power at points along the ray $\alpha \beta_2^*$. The power envelope gives the maximum power attainable by an invariant test for each set of β_2 and σ^2 values under H_a . If for an arbitrary choice of β_2^* value, the power of the test is very close to the power envelope, then there is little to be gained by trying to find a more optimal β_2^* value. In such cases the simplicity of the additive t -test makes it a very attractive test.

3. An empirical power comparison

In order to assess the usefulness of the additive t -test, we compared its empirical power in a range of circumstances with the power envelope, the power of the F -test and, in the case of two restrictions, the power of the exact likelihood ratio (LR) test. The F -test rejects H_0 for large values of

$$F = \{y'(\bar{P}_1 - \bar{P})y/j\} / \{y'\bar{P}y/(n-k)\},$$

where

$$\bar{P} = I_n - X(X'X)^{-1}X'.$$

Under (2.1) and (2.2), F follows a non-central F -distribution with j and $n - k$ degrees of freedom and non-centrality parameter

$$\theta = \beta_2' X_2' \bar{P}_1 X_2 \beta_2 / \sigma^2.$$

3.1. The experiment

In our experiment, the power of the F -test was computed for given X , β_2 , and σ using eq. (2) from Tiku (1967) and the IMSL subroutine MDBETA to evaluate Pearson's incomplete beta-function. The power of the additive t -test was calculated using Bulgren and Amos' (1968, pp. 1016–1018) algorithm for computing values of the doubly non-central t -distribution function.⁴ The LR test for the case of two restrictions was computed using the expressions given in Hillier (1986, sect. 4.4), together with the IMSL subroutines MDBETA and DGAMMA. The exact critical value is the solution to Hillier's (1986) eq. (7) and was found by iteration correct to six decimal places using subroutine MDFRE to invert the density of F .

The powers of the tests at the five percent significance level were calculated against all possible β_2 vectors whose components take any combination of the values $-0.5, 0.0, 0.5, 1.0, 2.0$. Although the additive t -test assumes the β_2 components are non-negative, the -0.5 values were included in order to assess the relative merits of the test when this assumption is violated. A value of σ was chosen individually for each X_2 matrix so that for the purposes of the comparison, the overall level of the powers is neither too high nor too low; i.e., so we are looking at interesting parts of the power curves. The X matrices used in the comparison were as follows:

- $X1$ ($n = 20, 60$; $k = 5$): A constant dummy, the quarterly seasonally adjusted Australian household disposable income and private consumption expenditure series, commencing 1967(2), and these two series lagged one quarter.
- $X2$ ($n = 20, 60$; $k = 6$): A constant dummy, three quarterly seasonal dummy variables, the quarterly Australian consumers' price index commencing 1959(1) and the same index lagged one quarter.

⁴Our version of this algorithm enabled us to reproduce Bulgren and Amos' table of doubly non-central t -distribution function values exactly to six decimal places.

- X3* ($n = 20, 60$; $k = 5$): A constant dummy, quarterly Australian private capital movements and quarterly Australian Government capital movements series commencing 1968(1) plus these two series lagged one quarter.
- X4* ($n = 20, 60$; $k = 6$): A constant dummy, three quarterly seasonal dummy variables, quarterly Australian retail trade and the latter series lagged one quarter.
- X5* ($n = 32, 64$; $k = 4$): Australian cross-section data classified according to eight categories of sex/marital status and eight categories of age. The variables are a constant dummy, population, number of households whose head belongs to the given population category and household headship ratios⁵ for 1961.
- X6* ($n = 20, 60$; $k = 5$): A constant dummy, two regressors of independent drawings from the uniform distribution with range (0,20) and two regressors of independent drawings from the log-normal distribution with a coefficient of variation of one.

These X matrices were chosen to represent a range of non-seasonal, seasonal, and cross-sectional data. The following fifteen X_2 matrices and associated σ values were used in the empirical power comparison:

- X1* (i) All non-constant regressors ($j = 4$) with $\sigma = 4$ for $n = 20$ and $\sigma = 30$ for $n = 60$.
 (ii) The two lagged regressors ($j = 2$) with $\sigma = 0.25$ for $n = 20$ and $\sigma = 0.35$ for $n = 60$.
 (iii) Current and lagged private consumption expenditure ($j = 2$) with $\sigma = 0.15$ for $n = 20$ and $\sigma = 0.7$ for $n = 60$.
- X2* (i) The three seasonal dummy variables ($j = 3$) with $\sigma = 2$ for $n = 20$ and $\sigma = 2.5$ for $n = 60$.
 (ii) Current and lagged Australian consumers' price index ($j = 2$) with $\sigma = 20$ for $n = 20$ and $\sigma = 125$ for $n = 60$.
- X3* (i) All non-constant regressors ($j = 4$) with $\sigma = 1000$ for $n = 20$ and $\sigma = 8000$ for $n = 60$.
 (ii) The two lagged regressors ($j = 2$) with $\sigma = 500$ for $n = 20$ and $\sigma = 2500$ for $n = 60$.
 (iii) Current and lagged quarterly Australian Government capital movements ($j = 2$) with $\sigma = 300$ for $n = 20$ and $\sigma = 1200$ for $n = 60$.
- X4* (i) The three quarterly seasonal dummy variables ($j = 3$) with $\sigma = 2$ for $n = 20$ and $\sigma = 2.5$ for $n = 60$.

⁵The household headship ratio is the proportion of people in any given population category who are heads of households. See Williams and Sams (1981) for further details of this data set.

- (ii) Current and lagged quarterly Australian retail trade ($j = 2$) with $\sigma = 10$ for $n = 20$ and $\sigma = 50$ for $n = 60$.
- X5 (i) All non-constant regressors ($j = 3$) with $\sigma = 12$ for $n = 32, 64$.
- (ii) Population and number of households ($j = 2$) with $\sigma = 5$ for $n = 32, 64$.
- X6 (i) All non-constant regressors ($j = 4$) with $\sigma = 40$ for $n = 20, 60$.
- (ii) One uniform and two lognormal regressors ($j = 3$) with $\sigma = 25$ for $n = 20, 60$.
- (iii) The two lognormal regressors ($j = 2$) with $\sigma = 5$ for $n = 20, 60$.

3.2. Results

The results of the power comparisons are summarised in tables 1–3, while some selected powers for $X1$, $X2$, and $X5$ are presented in tables 4–6. Because the powers were evaluated at all β_2 vectors whose components take any combination of the values $-0.5, 0.0, 0.5, 1.0, 2.0$, it follows that for $j = 2$ (3, 4), the tests are compared at 24 (124, 624) β_2 vectors excluding the origin, of which only 15 (63, 255) obey the constraint $\beta_2 \geq 0$ and 9 (27, 81) have only strictly positive components. For this reason, the summary statistics given in tables 1–3 are presented for all β_2 vectors, for $\beta_2 \geq 0$ and for β_2 vectors with strictly positive components. The exception is the results for the LR test which can only be given for $\beta_2 \geq 0$ and $\beta_2 > 0$ as the formulae in Hillier (1986) are invalid if any element of β_2 is negative.

In tables 1 and 2 we examine the power of the additive t , F and LR tests in comparison to the maximum attainable power as given by the power envelope. The most striking feature is the very high proportion of times that the additive t -test has power close to the power envelope. In particular for those vectors satisfying $\beta_2 > 0$ there are 9 (including 2 cases for which as many as 4 joint restrictions are being tested) out of a total of 30 cases for which the power is always 95% or more of the power envelope. Indeed, reading from table 2, amongst these cases the greatest difference from the power envelope is only 0.037, indicating that the room for improvement in power is only very small. Taking a snapshot from the total view, when $\beta_2 > 0$ we find the proportion lying within the above bound is 71% for the additive t -test which compares favourably with the LR test with only 56.3%. In comparison the F -test performs weakly, almost never reaching close to the power envelope, even in the most advantageous testing situation its power still ranges up to 0.203 from the power envelope.

In table 3 comparisons between tests are made. For $\beta_2 > 0$ and $\beta_2 \geq 0$ we find in all but 14 of the 60 cases that the additive t -test dominates the F -test. But of these 14 cases the additive t -test is always proportionally superior to the F -test. Generally speaking, the relative power of the F -test is at its best when

Table 1
Proportion of outcomes within 95% of the power envelope.

Test	n, j	Additive t -test			F -test			LR-test	
		All cases	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$	All cases	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$
X_2	20, 4	0.917	1.000	1.000	0.006	0	0		
	60, 4	0.923	1.000	1.000	0.013	0	0		
$X1$ (i)	20, 2	0.542	0.867	1.000	0	0	0	0.733	0.889
	60, 2	0.375	0.600	0.889	0	0	0	0.200	0.333
$X1$ (iii)	20, 2	0.292	0.467	0.778	0	0	0	0.067	0.111
	60, 2	0.750	1.000	1.000	0	0	0	1.000	1.000
$X2$ (i)	20, 3	0.024	0.048	0.111	0	0	0		
	60, 3	0.024	0.048	0.111	0	0	0		
$X2$ (ii)	20, 2	0.792	1.000	1.000	0	0	0	1.000	1.000
	60, 2	0.833	1.000	1.000	0	0	0	1.000	1.000
$X3$ (i)	20, 4	0.139	0.306	0.630	0	0	0		
	60, 4	0.290	0.486	0.889	0	0	0		
$X3$ (ii)	20, 2	0.208	0.333	0.556	0	0	0	0.133	0.222
	60, 2	0.500	0.733	0.889	0	0	0	0.267	0.333
$X3$ (iii)	20, 2	0.250	0.400	0.667	0	0	0	0	0
	60, 2	0.375	0.600	1.000	0	0	0	0.200	0.333
$X4$ (i)	20, 3	0.073	0.143	0.222	0	0	0		
	60, 3	0.032	0.063	0.148	0	0	0		
$X4$ (ii)	20, 2	0.792	1.000	1.000	0	0	0	1.000	1.000
	60, 2	0.792	1.000	1.000	0	0	0	1.000	1.000
$X5$ (i)	32, 3	0.556	0.905	1.000	0	0	0		
	64, 3	0.363	0.603	0.889	0	0	0		
$X5$ (ii)	32, 2	0.583	0.867	1.000	0	0	0	0.467	0.556
	64, 2	0.542	0.800	1.000	0	0	0	0.667	0.889
$X6$ (i)	20, 4	0.152	0.263	0.494	0	0	0		
	60, 4	0.146	0.255	0.481	0	0	0		
$X6$ (ii)	20, 3	0.427	0.651	0.815	0	0	0		
	60, 3	0.444	0.651	0.815	0	0	0		
$X6$ (iii)	20, 2	0.208	0.333	0.556	0	0	0	0	0
	60, 2	0.208	0.333	0.556	0.042	0.067	0.111	0.200	0.333
Total		0.398	0.532	0.710	0.000	0.000	0.000	0.496	0.563

all components of β_2 are negative. Typical situations in which the F -test is more powerful than the additive t -test are when one or more components of β_2 are negative and other components are zero or close to zero, although occasionally one component may be large while the others are negative or close to zero. In the four cases involving seasonal dummy variables, the F -test is sometimes more powerful than the additive t -test for β_2 with strictly positive

Table 2
Maximum power difference from the power envelope.

Test	n, j	Additive t -test			LR-test		F -test		
		All cases	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$	All cases	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$
$X1(i)$	20, 4	0.117	0.001	0.000			0.391	0.391	0.391
	60, 4	0.221	0.000	0.000			0.350	0.350	0.350
$X1(ii)$	20, 2	0.120	0.018	0.009	0.034	0.029	0.266	0.266	0.266
	60, 2	0.265	0.145	0.067	0.078	0.062	0.235	0.235	0.235
$X1(iii)$	20, 2	0.089	0.048	0.025	0.057	0.047	0.235	0.235	0.235
	60, 2	0.231	0.008	0.003	0.009	0.005	0.234	0.234	0.233
$X2(i)$	20, 3	0.642	0.454	0.269			0.352	0.346	0.325
	60, 3	0.842	0.643	0.385			0.301	0.301	0.301
$X2(ii)$	20, 2	0.105	0.001	0.001	0.002	0.001	0.261	0.261	0.261
	60, 2	0.219	0.000	0.000	0.000	0.000	0.234	0.234	0.234
$X3(i)$	20, 4	0.143	0.085	0.044			0.402	0.402	0.402
	60, 4	0.192	0.078	0.035			0.350	0.350	0.350
$X3(ii)$	20, 2	0.251	0.139	0.069	0.142	0.125	0.266	0.266	0.266
	60, 2	0.403	0.162	0.065	0.133	0.117	0.233	0.231	0.230
$X3(iii)$	20, 2	0.164	0.101	0.054	0.089	0.072	0.262	0.262	0.262
	60, 2	0.229	0.095	0.037	0.065	0.040	0.235	0.235	0.229
$X4(i)$	20, 3	0.623	0.444	0.268			0.349	0.321	0.291
	60, 3	0.816	0.638	0.408			0.301	0.301	0.301
$X4(ii)$	20, 2	0.075	0.001	0.001	0.001	0.001	0.203	0.203	0.203
	60, 2	0.206	0.002	0.001	0.003	0.002	0.235	0.235	0.235
$X5(i)$	32, 3	0.183	0.028	0.012			0.318	0.318	0.318
	64, 3	0.217	0.114	0.048			0.298	0.298	0.297
$X5(ii)$	32, 2	0.111	0.020	0.010	0.046	0.040	0.245	0.245	0.245
	64, 2	0.210	0.041	0.018	0.037	0.030	0.234	0.234	0.234
$X6(i)$	20, 4	0.218	0.137	0.074			0.385	0.385	0.385
	60, 4	0.464	0.273	0.124			0.350	0.350	0.349
$X6(ii)$	20, 3	0.144	0.057	0.025			0.345	0.345	0.345
	60, 3	0.296	0.122	0.052			0.299	0.299	0.299
$X6(iii)$	20, 2	0.318	0.199	0.103	0.117	0.089	0.266	0.266	0.266
	60, 2	0.519	0.279	0.106	0.100	0.068	0.235	0.235	0.235

components when one or perhaps two of these components is large and the remaining components are close to zero. For two joint restrictions we find the additive t -test dominating the LR test in 11 out of the 32 cases examined, while in the remaining 21 experiments the additive t -test is proportionally superior in 20 and marginally inferior in 1 case only. Typical instances in which the power of the LR test is superior to that of the additive t -test are confined mainly to when β_2 spanned the boundaries of the positive orthant;

Table 3
Further summary of the results of the empirical power comparison.

X_2	n, j	Proportion of times additive t -test more powerful than F -test			Proportion of times additive t -test more powerful than LR-test		Proportion of times LR-test more powerful than F -test	
		All cases	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$	$\beta_{2i} \geq 0$	$\beta_{2i} > 0$
$X1(i)$	20, 4	0.920	1.000	1.000				
	60, 4	0.920	1.000	1.000				
$X1(ii)$	20, 2	0.792	1.000	1.000	0.733	0.889	1.000	1.000
	60, 2	0.750	1.000	1.000	0.733	0.889	1.000	1.000
$X1(iii)$	20, 2	0.792	1.000	1.000	0.733	0.889	1.000	1.000
	60, 2	0.833	1.000	1.000	1.000	1.000	1.000	1.000
$X2(i)$	20, 3	0.331	0.651	0.963				
	60, 3	0.315	0.806	0.889				
$X2(ii)$	20, 2	0.833	1.000	1.000	1.000	1.000	1.000	1.000
	60, 2	0.833	1.000	1.000	1.000	1.000	1.000	1.000
$X3(i)$	20, 4	0.838	1.000	1.000				
	60, 4	0.824	1.000	1.000				
$X3(ii)$	20, 2	0.667	0.867	1.000	0.667	0.778	1.000	1.000
	60, 2	0.625	0.800	1.000	0.773	0.889	1.000	1.000
$X3(iii)$	20, 2	0.750	1.000	1.000	0.667	0.889	1.000	1.000
	60, 2	0.792	1.000	1.000	0.733	1.000	1.000	1.000
$X4(i)$	20, 3	0.427	0.587	0.815				
	60, 3	0.202	0.742	0.741				
$X4(ii)$	20, 2	0.833	1.000	1.000	1.000	1.000	1.000	1.000
	60, 2	0.833	1.000	1.000	1.000	1.000	1.000	1.000
$X5(i)$	32, 3	0.831	1.000	1.000				
	64, 3	0.790	1.000	1.000				
$X5(ii)$	32, 2	0.750	1.000	1.000	0.733	0.889	1.000	1.000
	64, 2	0.750	1.000	1.000	0.733	0.889	1.000	1.000
$X6(i)$	20, 4	0.798	1.000	1.000				
	60, 4	0.623	0.984	1.000				
$X6(ii)$	20, 3	0.742	0.984	1.000				
	60, 3	0.702	0.921	1.000				
$X6(iii)$	20, 2	0.625	1.000	1.000	0.600	0.889	1.000	1.000
	60, 2	0.542	0.867	1.000	0.467	0.778	1.000	1.000

for a specific design, the powers given in table 4 exhibit this pattern. The LR test dominates the F -test in all cases examined.

Finally, tables 5 and 6 give sets of powers that highlight weak and strong performances, respectively, of the additive t -test. In table 5, a case in which the tested regressors are orthogonal, the powers of both the additive t - and F -tests are significantly below the power envelope. Nevertheless, confining our

Table 4

Calculated powers of the power envelope, additive t , LR and F -tests for XI with X_2 comprising current and lagged consumption expenditure ($j = 2$).^a

$\beta_{21} =$	-0.5	0	0.5	1	2
$n = 20$					
$\beta_{22} = 2$	0.176	0.196	0.243	0.313	0.502
	0.095	0.148	0.218	0.303	0.502
	—	0.162	0.218	0.288	0.461
	0.086	0.094	0.116	0.151	0.267
1	0.102	0.106	0.143	0.206	0.391
	0.054	0.090	0.140	0.206	0.381
	—	0.093	0.135	0.191	0.349
	0.060	0.061	0.073	0.099	0.195
0.5	0.085	0.074	0.109	0.168	0.346
	0.040	0.068	0.109	0.165	0.323
	—	0.069	0.105	0.153	0.299
	0.055	0.053	0.062	0.083	0.169
0	0.087	0.050	0.087	0.143	0.311
	0.028	0.050	0.083	0.130	0.268
	—	0.050	0.079	0.121	0.254
	0.056	0.050	0.056	0.073	0.150
-0.5	0.109	0.074	0.085	0.131	0.286
	0.020	0.036	0.062	0.099	0.218
	—	—	—	—	—
	0.062	0.053	0.055	0.069	0.137
$n = 60$					
$\beta_{22} = 2$	0.485	0.645	0.785	0.887	0.980
	0.472	0.639	0.782	0.886	0.980
	—	0.636	0.780	0.884	0.979
	0.274	0.411	0.564	0.712	0.914
1	0.149	0.262	0.414	0.582	0.855
	0.142	0.260	0.413	0.582	0.855
	—	0.258	0.411	0.579	0.853
	0.078	0.132	0.223	0.352	0.662
0.5	0.067	0.127	0.236	0.385	0.714
	0.058	0.126	0.236	0.384	0.711
	—	0.126	0.235	0.383	0.709
	0.052	0.069	0.118	0.204	0.481
0	0.112	0.050	0.112	0.216	0.530
	0.019	0.050	0.111	0.213	0.522
	—	0.050	0.111	0.213	0.522
	0.064	0.050	0.064	0.108	0.309
-0.5	0.236	0.127	0.067	0.106	0.343
	0.005	0.016	0.043	0.098	0.328
	—	—	—	—	—
	0.118	0.069	0.052	0.062	0.178

^a The first power in each cell is for the power envelope, the second the additive t -test, the third the LR-test, and the fourth the F -test.

Table 5

Calculated powers of the power envelope, additive t - and F -tests for X_2 with $n = 60$ and X_2 comprising seasonal dummy variables ($j = 3$).^a

$\beta_{22} =$	-0.5	0	0.5	1	2
$\beta_{21} = 1$					
$\beta_{23} = 2$	0.900	0.812	0.731	0.698	0.813
	0.255	0.346	0.443	0.539	0.708
	0.663	0.525	0.430	0.397	0.527
1	0.625	0.447	0.350	0.373	0.696
	0.148	0.213	0.290	0.373	0.540
	0.331	0.206	0.154	0.166	0.395
0.5	0.514	0.344	0.282	0.351	0.730
	0.104	0.156	0.219	0.290	0.445
	0.249	0.151	0.122	0.155	0.429
0	0.495	0.365	0.348	0.452	0.813
	0.069	0.107	0.156	0.214	0.349
	0.237	0.162	0.153	0.210	0.527
-0.5	0.578	0.500	0.521	0.632	0.901
	0.043	0.069	0.105	0.148	0.258
	0.294	0.240	0.254	0.337	0.666
$\beta_{21} = 0.5$					
$\beta_{23} = 2$	0.887	0.806	0.742	0.731	0.859
	0.192	0.270	0.357	0.448	0.621
	0.641	0.518	0.443	0.431	0.593
1	0.522	0.352	0.287	0.353	0.728
	0.106	0.158	0.222	0.294	0.449
	0.255	0.155	0.124	0.155	0.427
0.5	0.348	0.187	0.163	0.286	0.738
	0.073	0.112	0.163	0.222	0.359
	0.153	0.084	0.076	0.124	0.438
0	0.280	0.160	0.189	0.353	0.803
	0.047	0.075	0.113	0.159	0.272
	0.121	0.075	0.085	0.155	0.514
-0.5	0.346	0.284	0.352	0.525	0.887
	0.029	0.047	0.073	0.106	0.194
	0.152	0.123	0.155	0.256	0.640

^a The first entry in each cell is the power envelope, the second the additive t -test, and the third the F -test.

attention to strictly positive components in β_2 the additive t -test has superior power to that of the F -test at 16 points out of 18. In contrast the powers of the additive t -test given in table 6 are very close to the power envelope at all points examined including points outside the positive orthant. Indeed the relative power of the F -test is so poor even along the boundary and for negative β_2 components that it has barely more power than the additive t -test at the extreme point, $\beta_2 = (0.5, -0.5, -0.5)'$.

Table 6

Calculated powers of the power envelope, additive t - and F -tests for X_5 with $n = 32$ and X_2 comprising all non-constant regressors ($j = 3$).^a

$\beta_{22} =$	-0.5	0	0.5	1	2
$\beta_{21} = 1$					
$\beta_{23} = 2$	0.157	0.244	0.358	0.490	0.744
	0.149	0.241	0.357	0.488	0.740
	0.073	0.103	0.153	0.224	0.427
1	0.142	0.221	0.329	0.457	0.716
	0.131	0.217	0.328	0.457	0.714
	0.068	0.095	0.139	0.205	0.399
0.5	0.137	0.211	0.316	0.442	0.702
	0.123	0.205	0.314	0.441	0.701
	0.067	0.091	0.133	0.196	0.385
0	0.133	0.203	0.304	0.427	0.689
	0.115	0.194	0.300	0.426	0.687
	0.066	0.088	0.128	0.188	0.372
-0.5	0.131	0.196	0.292	0.413	0.675
	0.107	0.183	0.286	0.410	0.672
	0.065	0.086	0.123	0.180	0.360
$\beta_{21} = 0.5$					
$\beta_{23} = 2$	0.084	0.140	0.228	0.343	0.612
	0.076	0.137	0.224	0.337	0.599
	0.054	0.068	0.097	0.146	0.308
1	0.074	0.121	0.202	0.311	0.577
	0.065	0.120	0.201	0.309	0.569
	0.052	0.062	0.088	0.131	0.282
0.5	0.072	0.114	0.190	0.296	0.560
	0.060	0.112	0.190	0.295	0.553
	0.052	0.060	0.084	0.125	0.270
0	0.073	0.108	0.180	0.282	0.544
	0.056	0.105	0.180	0.281	0.538
	0.052	0.059	0.080	0.119	0.259
-0.5	0.077	0.104	0.172	0.270	0.528
	0.052	0.098	0.170	0.268	0.522
	0.053	0.058	0.077	0.114	0.248

^aSee the footnote to table 5.

It would seem from the results reported that an important determinant of the relative power of the tests is the degree of collinearity between pairs of columns of X_2 . Clearly the additive t -test is least dominant when the columns of X_2 are comprised of three quarterly seasonal dummy variables which are orthogonal. The additive t -test seems to perform better when the columns of X_2 are highly collinear. Intuitively, one might expect that given j and n , the relative performance of the additive t -test is determined in part by the relative

size of the subspace

$$S = \{ X_2 \beta_2 : \beta_2 \geq 0 \}.$$

When the X_2 regressors are close together in R^n space, S is relatively small and it is not surprising that the additive t -test which is UMPI along

$$\{ \alpha X_2 \beta_2^* : \alpha > 0, \beta_2^* = (1, \dots, 1)' \}$$

dominates over the subspace S . Also our experiments indicate that the test is reasonably insensitive to the choice of β_2^* in such cases. On the other hand, if the X_2 regressors are orthogonal, then S is relatively large in size and it is less likely that the additive t -test can outperform, for instance, the F -test over all points in the subspace S . Furthermore, the choice of β_2^* plays a much more critical role.

4. Concluding remarks

This paper proposes a simple test of linear restrictions in the linear regression model against one-sided alternatives. As well as being UMPI in the neighbourhood of a predetermined subspace of the parameter space, an empirical power comparison indicates that the new test can have power very close to the power envelope over a wide range of parameter values. When testing whether a subset of regression coefficients are jointly zero, the power advantage of the new test appears to be greatest when the associated regressors are highly collinear. It therefore would seem to suit especially the joint testing of coefficients of lagged values of a given variable.

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